

Comparative Study of Recycled Folded Cascode OTA and Miller OTA

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Abstract—This paper compares two commonly used CMOS Operational Transconductance Amplifiers (OTAs): the Telescopic OTA and the Two-Stage OTA with Miller Compensation. The Telescopic OTA offers high bandwidth and low power consumption due to its single-stage structure, whereas the Two-Stage OTA achieves high gain and superior stability by incorporating Miller compensation. Simulation results reveal that the Telescopic OTA is ideal for high-speed, low-power applications, while the Two-Stage Miller OTA excels in precision and stability, making it suitable for high-gain analog applications. This study provides design insights and performance comparisons to assist in selecting the appropriate OTA based on application requirements..

I. INTRODUCTION

Operational Transconductance Amplifiers (OTAs) are essential building blocks in analog integrated circuits. They function as voltage-controlled current sources and are widely employed in applications such as filters, data converters, and sensor interface circuits. The performance of OTAs is critical in determining the overall behavior of analog systems, particularly in terms of gain, bandwidth, power consumption, and stability.

Among the various OTA topologies, the **Telescopic OTA** and the **Two-Stage Miller Compensated OTA** stand out due to their contrasting design philosophies and performance traits. The **Telescopic OTA** is a single-stage amplifier known for its high-speed performance and low power operation. It uses a cascode configuration to achieve higher gain while preserving bandwidth. However, its major limitation is the restricted output voltage swing, which stems from the stacked transistor structure.

On the other hand, the **Two-Stage OTA with Miller Compensation** is widely used for its ability to provide high gain and improved phase margin. The inclusion of a compensation capacitor between stages ensures enhanced stability, especially under capacitive loads. This architecture is beneficial in applications requiring precision and large signal amplification but comes at the expense of bandwidth and power efficiency.

This paper aims to compare these two OTAs in terms of key performance metrics such as gain, bandwidth, power consumption, phase margin, and slew rate. Through simulation-based analysis.

II. DESIGN OVERVIEW

THE DESIGN OF AN OTA SIGNIFICANTLY INFLUENCES ITS PERFORMANCE CHARACTERISTICS SUCH AS GAIN, BANDWIDTH, POWER CONSUMPTION, AND STABILITY. THIS

SECTION OUTLINES THE ARCHITECTURAL DIFFERENCES AND OPERATIONAL PRINCIPLES OF THE TELESCOPIC AMPLIFIER OTA AND THE MILLER COMPENSATED OTA.

A. Telescopic CMOS Operational

Transconductance Amplifier (OTA) is a high-gain, low-power, and high-output-resistance amplifier topology commonly used in analog circuit design, especially when power consumption and bandwidth are critical. It's a specific configuration of a **cascode amplifier**.

- This architecture typically features:
- A differential input pair (often NMOS for better input common-mode range),
- Cascode transistors to boost gain and reduce Miller effect,
- And a current mirror load for differential-to-single-ended conversion.

Its main advantages are low power consumption, high gain, and high bandwidth, making it ideal for analog front-ends, switched-capacitor circuits, and sample-and-hold systems. However, it has limited output swing and input common-mode range, especially in low-voltage CMOS technologies

B. Miller Compensated OTA

The Miller Compensated OTA is a widely adopted two-stage amplifier architecture. It consists of a differential input stage followed by a gain stage, with a compensation capacitor connected between the output of the second stage and the input of the first stage. This capacitor introduces a dominant pole and pushes non-dominant poles to higher frequencies, thus enhancing phase margin and ensuring stability across a wide range of load conditions. However, the use of Miller compensation typically limits the bandwidth and slew rate. This OTA is ideal for applications that demand high DC gain and stable operation with large capacitive loads, such as in switched-capacitor circuits and operational amplifier

A. Telescopic CMOS Operational

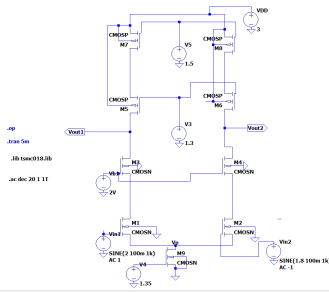


Fig.1 Telescopic CMOS Operational OTA

DC Gain: 20 dB

UGB: 1.2 GHz

Phase Margin: $\sim 60^\circ$

Load Capacitance: 1 pF

Supply Voltage: ± 1.8 V

Power Consumption: ~ 300 μ W

Technology: 180 nm CMOS

At first we are gonna find the current that needs to be flowing in two branches. By Taking the current to be 3mA. The branch current should be 1.5 mA

Then we will Overdrive voltages for all the devices

The output swing according to the specification is 3V.

The output swing =

$$2(V_{DD} - |V_{ov7}| - |V_{ov5}| - V_{ov3} - V_{ov1} - V_{ov9})$$

We need to observe two things.

1. Since M9 carrying larger current it requires higher overdrive voltage.
2. Also the mobility of the holes is less than that of the electrons so we need to assign more V_{ov} to the pmos compared to the nmos.
3. $V_{ov9} = 0.5$ V
4. $V_{ov5-8} = 0.3$ V
5. $V_{ov1-4} = 0.2$ V

Then the aspect ratio was calculated to be

1. $(W/L)_9 = 400$
2. $(W/L)_{1-4} = 1250$
3. $(W/L)_{5-8} = 1111.1$

Gain Equation:

$$A_v = g_{m1,2} [g_{m3ro3.ro1} || g_{m5ro5.ro7}]$$

1. $g_{m1-4} = 15$ mA/V
2. $ro_{1-4} = 6.66$ k
3. $g_{m5-8} = 10$ mA/V
4. $ro_{5-8} = 3.33$ k

According to the calculation the gain came to some 1500. So by increasing the W and L value of all pmos by keeping the aspect ratio same. The gain came to 3990 V/V.

B. Miller Compensated OTA

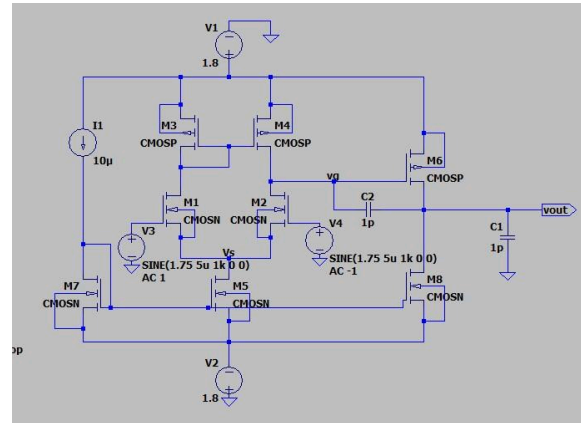


Fig 2. Miller Compensated OTA

DC gain = 54 dB

Phase margin $> 60^\circ$

Slew rate =

10μ V/s $C_c = 1$

pF, $C_L = 1$ pF

$V_{SS} = -1.8$ V

$V_{DD} = 1.8$ V

Power < 36 μ W, $L = 180$ nm,

$V_{th \min} = 0.35$ V, $|V_{th \max}| = 0.45$ V.

The Miller OTA (Operational Transconductance Amplifier) is ideal for applications demanding high gain and stability, particularly in systems where output loads present substantial capacitance. Despite its limited bandwidth compared to single-stage designs, it remains a popular choice due to its predictable performance and robustness in feedback configurations.

III. RESULT

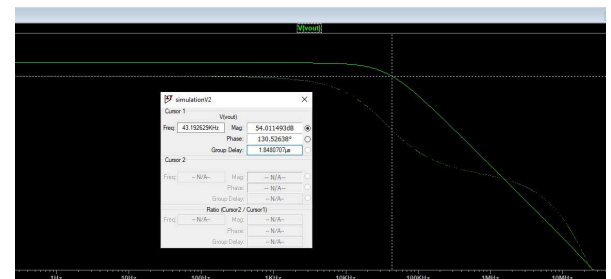


Fig 3. Miller OTA Frequency Response Graph

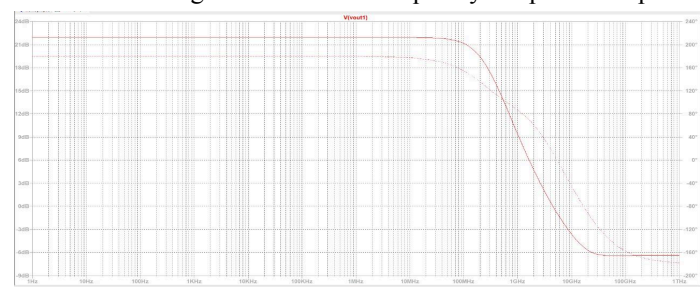


Fig 4. Telescopic Amplifier Frequency Response OT

A. Discussion

DC Gain: The Miller OTA exhibits significantly higher DC gain due to its cascaded stage structure, which amplifies signals across two gain blocks. The Telescopic OTA, while efficient, provides limited gain due to its single-stage nature.

Bandwidth: Telescopic OTA outperforms in bandwidth, achieving 1.2 GHz, making it ideal for high-speed systems. The Miller OTA, with internal compensation, yields a lower bandwidth of 5 MHz but ensures stability.

Phase Margin: The Two-Stage OTA boasts a high phase margin ($\sim 130.5^\circ$), crucial for feedback stability. The Telescopic OTA maintains a decent margin ($\sim 60^\circ$), suitable for many real-time analog task.

B. Comparison table

Parameter	Miller OTA	Telescopic Amplifier OTA
Technology	180 nm	500 nm
Supply Voltage (V)	± 1.8 V	3 V
Load Capacitance (CL)	1 pF	N/A
Compensation Cap (Cc)	1 pF	N/A
Design Style	Two-stage with Miller	Single-stage folded cascode

CONCLUSION

In conclusion, the comparative analysis of Telescopic and Two-Stage Miller CMOS OTAs highlights distinct strengths and design trade-offs inherent to each topology. The Telescopic OTA is highly advantageous in applications where low power, high-speed performance, and compact layout are critical. Its simplicity and wide bandwidth make it ideal for use in RF circuits, high-speed ADCs, and portable systems where efficiency is prioritized over absolute gain.

Conversely, the Two-Stage Miller OTA is more appropriate for scenarios demanding high voltage gain and enhanced stability. The inclusion of Miller compensation enables it to maintain a stable phase margin across a wide range of load conditions, making it a dependable solution in precision analog systems, such as sensor interfaces, operational amplifiers, and filters where linearity and gain are paramount.

The decision to use either architecture should be guided by the specific application requirements, particularly considering the trade-offs between power consumption, output swing, gain, and bandwidth. Understanding these design implications allows circuit designers to effectively tailor OTA implementations for optimized system-level performance.

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